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● *Clinical Original Contribution*

HIGH INTENSITY 125-IODINE (^{125}I) PLAQUE TREATMENT OF UVEAL MELANOMA

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Purpose: Episcleral ^{125}I plaque therapy of uveal melanoma is an important treatment modality to control tumor, salvage the globe, and potentially preserve vision. We retrospectively analyzed our experience in 239 patients to assess treatment outcome with this technique.

Methods and Materials: Between 1983 and 1990, 239 uveal melanoma patients were treated with ^{125}I plaques at the University of California, San Francisco. High intensity ^{125}I seeds in the range of 3–20 mCi were used to give a minimum tumor dose of 70 Gy in 4 days. Initial mean tumor size was 10.9 mm × 9.2 mm × 5.5 mm with a range in tumor diameter from 4 to 18 mm and tumor height from 1.9 to 11.1 mm. Best corrected pre-treatment visual acuity was 20/200 or better in 92% of patients.

Results: Local tumor control was maintained in 91.7% of patients with a mean follow-up of 35.9 months; 19 patients had local tumor progression; mean time to progression was 27.3 mo (1.8 to 60.1 mo). Actuarial local control is 82% at 5 years. Multivariate analysis demonstrates significant correlation of local failure with larger maximum tumor diameter (p = 0.0008), closer proximity to the fovea (p = 0.0001), lower radiation dose (p = 0.0437), and smaller ultrasound height (p = 0.0034). The actuarial incidence of distant metastases is 12% at 5 years with multivariate analysis showing significant correlation only with maximum tumor diameter (p = 0.0064). Visual outcome is 20/200 or better in 58% of patients.

Conclusion: While the tumor control rates appear favorable, ocular morbidity is significant. A current randomized trial comparing ^{125}I plaque with Helium ion therapy is in progress with specific comparison of tumor control, survival, and visual outcome.

Iodine 125, Episcleral plaques, Uveal melanoma, Brachytherapy, Clinical trial, Control rates, Visual outcome.

INTRODUCTION

The optimum management of uveal melanoma is uncertain. Uveal melanoma is a rare tumor with approximately 1500 new cases diagnosed in the United States each year. Brachytherapy to treat these neoplasms was initially evaluated with the use of radon seeds (13) and later with episcleral ^{60}Co plaques (17, 18). Unfortunately, these high energy isotopes made it impossible to shield the ocular adnexa and consequently, although tumor control was excellent, the outcome was suboptimal with significant morbidity including vision loss and injury to the lacrimal apparatus and eye lids (4, 6). Renewed enthusiasm to explore brachytherapy techniques was kindled with the availability of lower energy isotopes such as ^{125}I that could be individually arrayed for a specific tumor geometry. Potential advantages of this technique included the ability to shield the ocular adnexa with a 3 mm gold foil and lowered radiation dose to the surgical team (12, 14, 16, 19).

At the University of California San Francisco (UCSF), an on-going program to explore the use of helium ions for the treatment of uveal melanoma was in place in 1982 with promising early results (3, 5, 15). A randomized trial comparing charged particle beams with ^{125}I plaques could reveal which, if either, was the superior therapy. In 1983, a Phase I/II trial was instituted to explore the use of ^{125}I at UCSF. In 1985, a prospective randomized trial to compare helium ions with ^{125}I was opened.

METHODS AND MATERIALS

Between 1983 and 1990, 239 patients with uveal melanoma were treated with ^{125}I episcleral plaques at UCSF. One hundred fifty-seven of these patients were previously

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